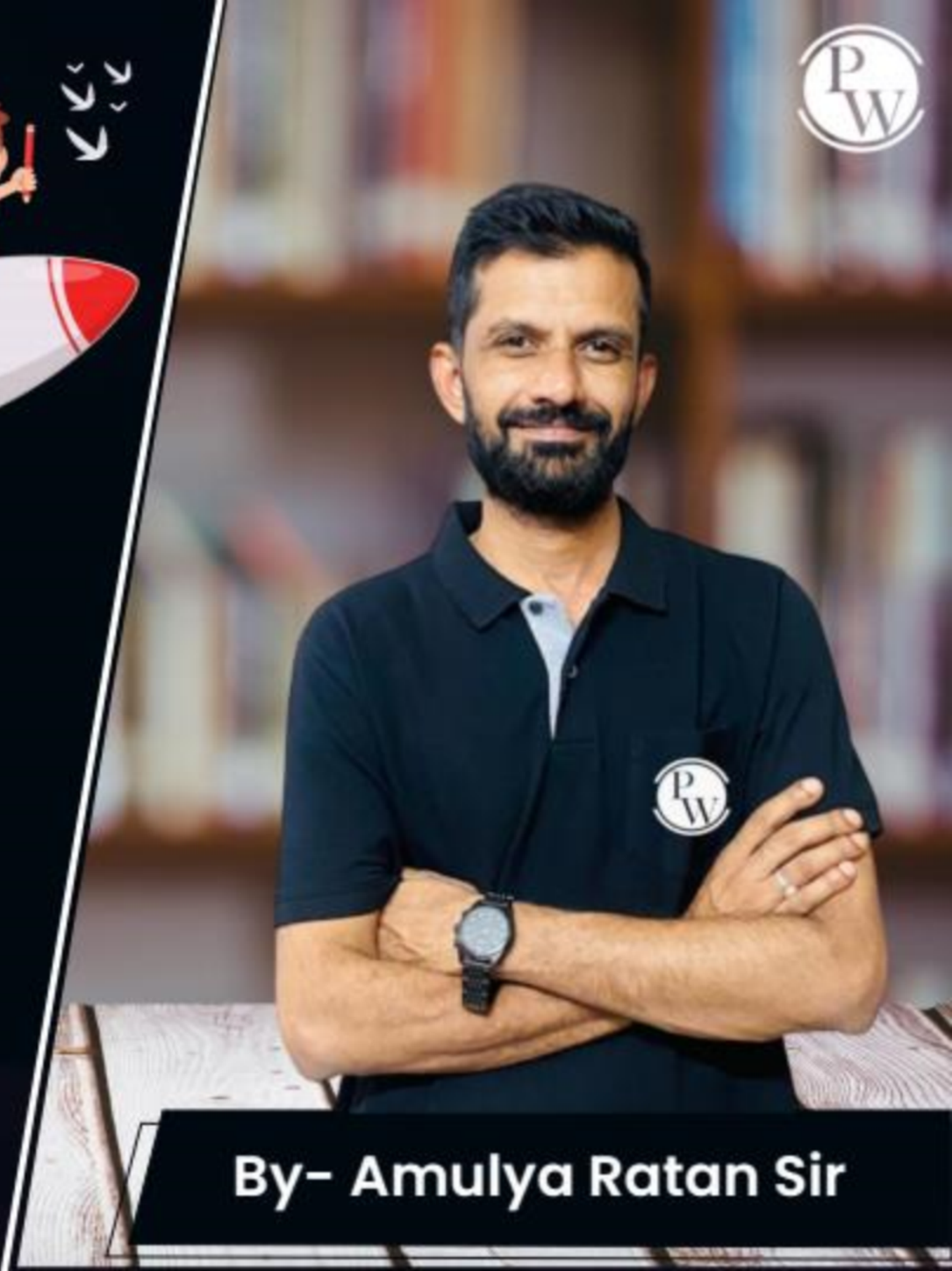
GATE ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 12



By- Amulya Ratan Sir

Recap of Previous Lecture



Topic Time and Work, Chain Rule

Topics to be Covered



Topic-1 Pipes and Cistern

[MCQ]

#Q. A Clock which gains 10 minutes in every one hour was set correct at 6 am. If that clock represents 1 pm the same day, what must be the correct time?

12pm

11:50

Assignment

C.W
1 hr
60 min

W.W
1 hr + 10 min
70 min

60 × 7/70 = ?

7 hr × 60

6 hr

[MCQ]

#Q. A Clock which gains 10 minutes in every one hour was set correct at 6 am. If the correct time is 12 pm the same day, what would be the time shown by that clock?

1 pm

[MCQ]

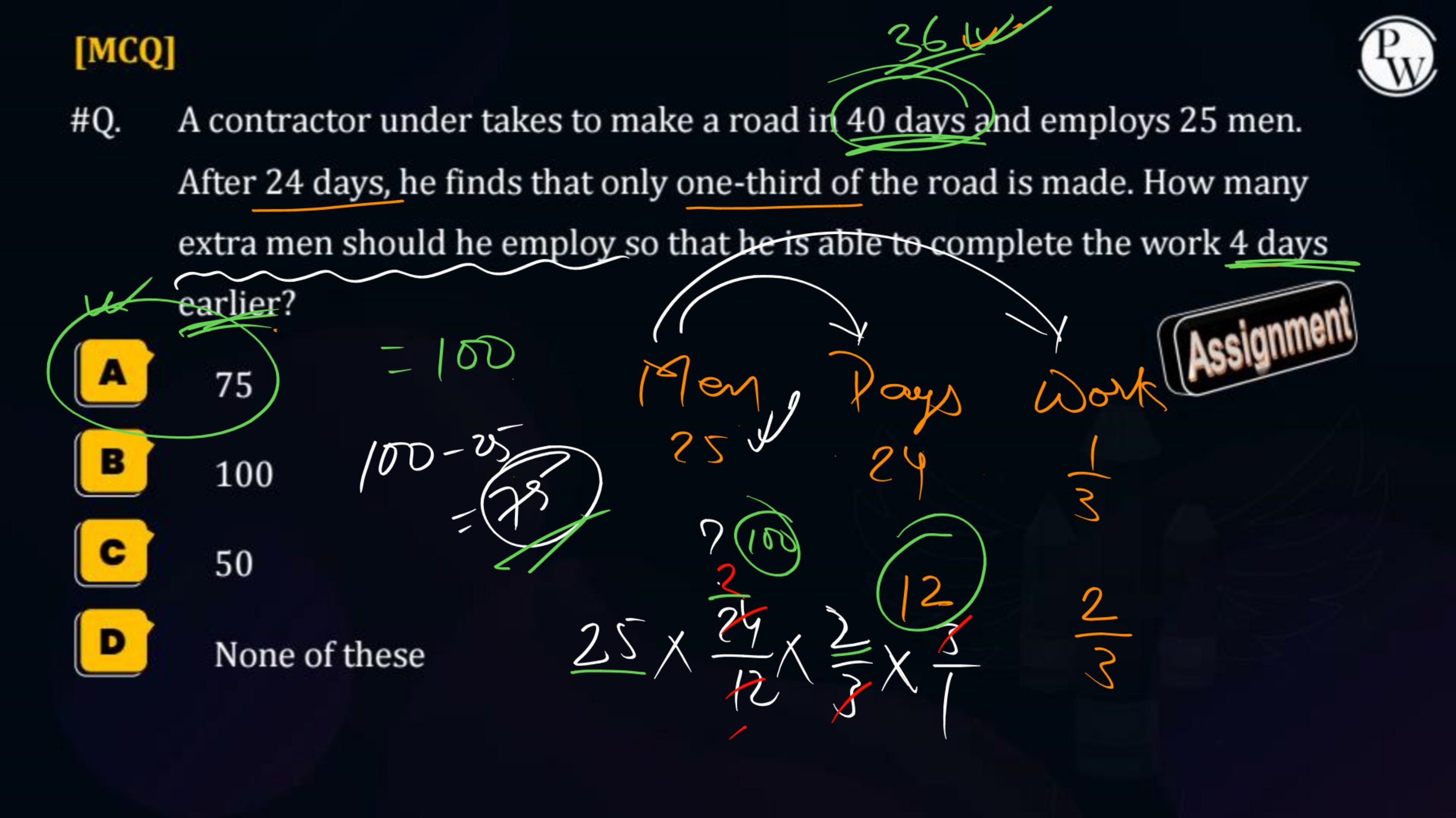
#Q. A contractor under takes to make a road in 40 days and employs 25 men. After 24 days, he finds that only one-third of the road is made. How many extra men should he employ so that he is able to complete the work 4 days earlier?

A. 75

B. 100

C. 50

D. None of these



[MCQ]

#Q. A can do a work in 60 days whereas B in 20 days. If they together took a contract of ₹32000, then what would be the share of A?

Assignment

A ₹ 8000

B ₹ 6000

C ₹ 1800

D ₹ 2040

$A:B = \frac{1}{60} : \frac{1}{20}$ (Efficiency)

$A:B = 1:3$

$\frac{1}{4} \times 32,000 = 8000$

Pipes & Cistern

↓

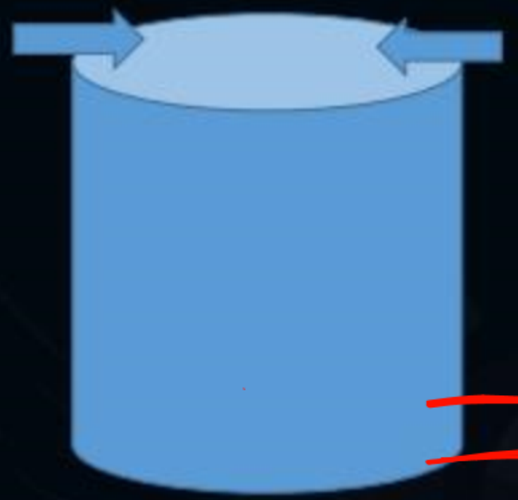
Tank

Time & Work

Pipes & Cistern

Empty (−) ✓
filling (+) ✓

Outlet ✓
Drain ✓
Leak ✓



$\frac{1}{6} + \frac{1}{5} - \frac{1}{4}$

$\frac{22-15}{60}$

$= \frac{7}{60}$

$\frac{60}{7} = 8\frac{4}{7}$ Lts

$A \& B = \frac{1}{5} + \frac{1}{6}$

$= \frac{11}{30}$

$\frac{30}{11} = 2\frac{8}{11}$ hrs

$\frac{1}{5}$

$\frac{1}{6}$

$= \frac{1}{4}$

9 Lts

[MCQ]

#Q. Two pipes A and B can fill a tank in 8 minutes and 12 minutes respectively.

Both the pipes are opened together and after 3 minutes, pipe A is turned off.

What is the total time required to fill the tank?

A. 3 minutes

B. 7.5 minutes

C. 10 minutes

D. 4.5 minutes

$A = \frac{1}{8}$ | $B = \frac{1}{12}$

$\frac{3}{8} + \frac{x}{12} = 1$

$\frac{9+2x}{24} = 1$

$\Rightarrow 9 + 2x = 24$

$2x = 15$

$x = \frac{15}{2} = 7.5$ min

1 unit $\left(\frac{1}{8}\right)$ = 12

$B = 12 \times \frac{1}{2} = 6$

$= \frac{15}{2}$

$= 7.5$

[MCQ]

#Q. Pipes A and B can fill a tank in 4 hours and 8 hours respectively. Pipe C can empty it in 16 hours. If all the three pipes are opened together, then how long will it take to fill the tank?

A. 4.5 hours

B. 2 hours

C. 6 hours

D. 3.2 hours

$A = \frac{1}{4} \quad B = \frac{1}{8} \quad C = \frac{-1}{16}$

$\frac{1}{4} + \frac{1}{8} - \frac{1}{16} = \frac{6-1}{16}$

$= \frac{5}{16} \Rightarrow \frac{16}{5}$

$= 3.2$

[MCQ]

#Q. A tank can be filled by 20 buckets each of capacity 13.5 litres. If the capacity of each bucket be 9 litres, how many buckets will fill the same tank?

A 30

B 25

C 20

D 15

Tank Capacity

$20 \times 13.5 = x \times 9$

$\frac{270}{9} = x$

30

1

Bucket	Capacity
20	13.5
More	9

$20 \times \frac{13.5}{9}$

[NAT]

#Q. A tank consist of three taps A, B and C. Tap A and tap B can fill the tank in 20hrs and 12 hrs respectively, whereas tap C can empty the tank in 18hrs. C was kept open whereas A and B are opened in alternate hours i.e. 1st hr A, 2nd hr B, 3rd hr A.....so on. In how many hours the tank would be filled?

$A = \frac{1}{20}$ | $B = \frac{1}{12}$ | $C = -\frac{1}{18}$

1 2 3 4 5 ... — $\frac{1}{45}$?

A & C | B & C | A & C | B & C | A & C

$8\frac{4}{5}$ hrs

89 hrs → $\frac{44}{45}$

1st hr: A & C $\Rightarrow \frac{1}{20} - \frac{1}{18} = \frac{-1}{180}$

2nd hr: B & C $\Rightarrow \frac{1}{12} - \frac{1}{18} = \frac{5}{180} = \frac{1}{36}$

1 work B & C = 36 hrs

$\frac{1}{45}$ " " = $36 \times \frac{1}{45}$ = 2nd hr

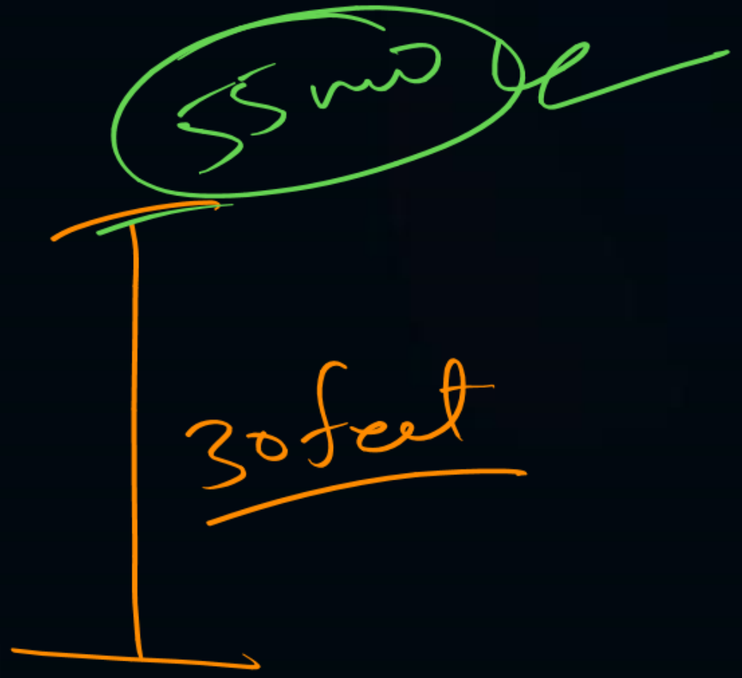
2 hrs

2nd & 3rd

$\frac{4}{5}$ hrs

$\frac{5}{180} - \frac{1}{180} = \frac{4}{180} = \frac{1}{45}$ (×44) → 2 hrs (×44)

$\frac{44}{45}$ → 88 hrs



1 min → +3 ft

2nd min → −2 ft

2 min ⇒ 1 feet

54 min ⇒ 27 ft.

53 min → +3

56 min → 28 ft

Time and Distance

km/hr $\xrightarrow{\times \frac{5}{18}}$ m/sec

m/sec $\xrightarrow{\times \frac{18}{5}}$ km/hr

- Average Speed
- Relative Speed

$$\text{Speed} = \frac{D}{T}$$

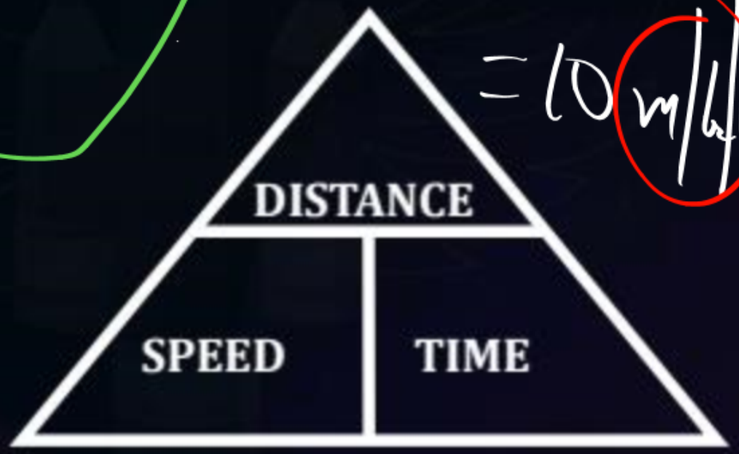
$$D = \text{Speed} \times \text{Time}$$

$$\text{Time} = \frac{D}{S}$$

$$\frac{\text{km}}{\text{hr}} = \frac{1000\text{m}}{60 \times 60\text{sec}} \quad (\times 5,\ \div 18)$$

Ex. $D = 400\text{m}$, $S = 40\text{km/hr}$

$$T = \frac{400\text{m}}{40\text{km/hr}} = 10\ \text{m/hr}$$



[MCQ]

#Q. If you travel from M to N at 40km/hr and N to M at 60 km/hr. What would be your average speed of the journey?

Assignment

[MCQ]

#Q. A man travels from Howrah to Kanpur in his car. If he covers $1/3^{rd}$ of the distance at 60 km/hr & remaining at 40 km/hr. Find the average speed of his journey.

Assignment

[MCQ]

#Q. A man travels ¼th of his journey at 20 km/hr, another ¼th at 30 km/hr & remaining at 60 km/hr. Find the average speed of his journey.

Assignment



2 mins Summary

Topic Pipes and Cistern, Basics of Time & Distance

THANK - YOU